

CAN-Based Engine Monitoring & Vehicle Control System

A High-Reliability Distributed Automotive Embedded System Built with NXP LPC2129 (ARM7TDMI-S) Microcontrollers, Embedded C, ISO 11898 CAN 2.0B Protocol, Non-Blocking Real-Time Supervision, DSP Signal Linearization, and Fault Recovery

System Architecture	3-Node Distributed CAN Bus Network (250 kbps Baud Rate)
Target Microcontroller	NXP LPC2129 (ARM7TDMI-S @ 60 MHz CCLK, 16/32-bit RISC)
Physical Communications	ISO 11898 CAN 2.0B Differential Bus (MCP2551 Transceivers, 120 Ω Termination)
Master Telemetry Cluster	20x4 Alphanumeric LCD (HD44780), DS18B20 1-Wire Digital Thermal Sensor
Body Control Subsystem	8-Level Bidirectional Power Window Motor Actuator & Discrete LED Array
ADAS Proximity Subsystem	Sharp GP2D12 IR Sensor with 20-Sample Trimmed Mean Filter & Hysteresis
Supervisory Safety System	16-Cycle Software Watchdog, Auto-Recovery & Dedicated Node Error Screens
Toolchain & Verification	Keil μ Vision 4/5 (ARM Compiler), Proteus VSM Automotive Simulation

Executive Summary: Modern vehicular electronic architectures necessitate robust, deterministic, and fault-tolerant inter-subsystem communications to manage real-time power delivery, passenger comfort, and advanced driver assistance systems (ADAS). This comprehensive engineering document details the design, algorithmic implementation, physical bus layout, and firmware architecture of a distributed three-node automotive control area network. The system partitions responsibilities between a Master Instrument Cluster ECU, a Body Control Power Window ECU, and a Rear Parking Radar ADAS ECU. All communication is arbitrated non-destructively over a dual-wire differential CAN 2.0B physical bus operating at 250 kbps with full software watchdog supervision and immediate error isolation.

1. System Architecture & Distributed CAN Bus Topology

In modern automotive engineering, centralized electronic architectures have been replaced by distributed Electronic Control Units (ECUs) interconnected via Controller Area Network (CAN) buses. This design eliminates heavy, cost-prohibitive point-to-point wiring harnesses, reduces electro-magnetic susceptibility, and enhances modular reliability. This project implements an automotive-grade distributed architecture consisting of three dedicated NXP LPC2129 ARM7 microcontrollers communicating over an ISO 11898 physical differential bus terminated with twin 120 Ω resistors.

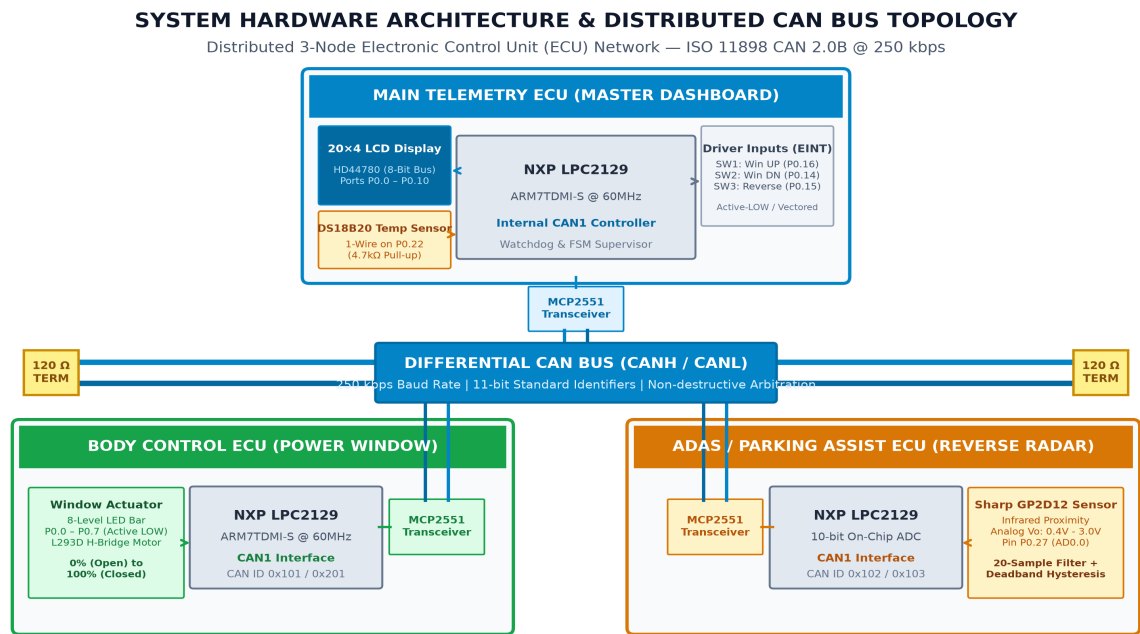


Figure 1: Distributed Hardware System Architecture and Physical CAN Bus Topology.

1.1 Node Functional Specialization

- 1. Master Instrument Cluster ECU (Main Node):** Positioned on the driver console, this node orchestrates overall system telemetry. It directly samples an engine block DS18B20 digital temperature sensor over a 1-Wire interface, monitors three hardware interrupt buttons (EINT0, EINT1, EINT2) for driver commands, drives a 20×4 alphanumeric LCD display using an 8-bit parallel bus, and executes supervisory software watchdogs to immediately detect the failure or disconnection of any peripheral ECU.
- 2. Body Control ECU (Window Node):** Manages power window actuation. It listens on CAN ID 0x101 for roll-up and roll-down commands, controls an 8-level discrete position indicator LED bar (Ports P0.0–P0.7) and L293D H-bridge motor driver, prevents limit-overtravel, and broadcasts state acknowledgement telemetry back to the master cluster via CAN ID 0x201.
- 3. ADAS Parking Assist ECU (Reverse Node):** Houses a Sharp GP2D12 infrared optical triangulation distance sensor connected to the LPC2129 on-chip 10-bit Successive Approximation ADC (Port P0.27 / AD0.0). Upon CAN enablement from the master cluster, this node executes a high-speed 7-stage Digital Signal Processing (DSP) pipeline to linearize sensor voltages, reject noise, and stream obstacle proximity telemetry at 10 Hz over CAN ID 0x103.

2. CAN 2.0B Physical Layer, Bit Timing & Signaling

The Controller Area Network protocol (ISO 11898) utilizes balanced differential signaling across two twisted wires: CANH and CANL. The differential voltage $V_{diff} = V_{CANH} - V_{CANL}$ provides exceptional common-mode noise rejection, which is mandatory in high-noise automotive environments plagued by inductive alternator whine, ignition sparks, and motor transient spikes.

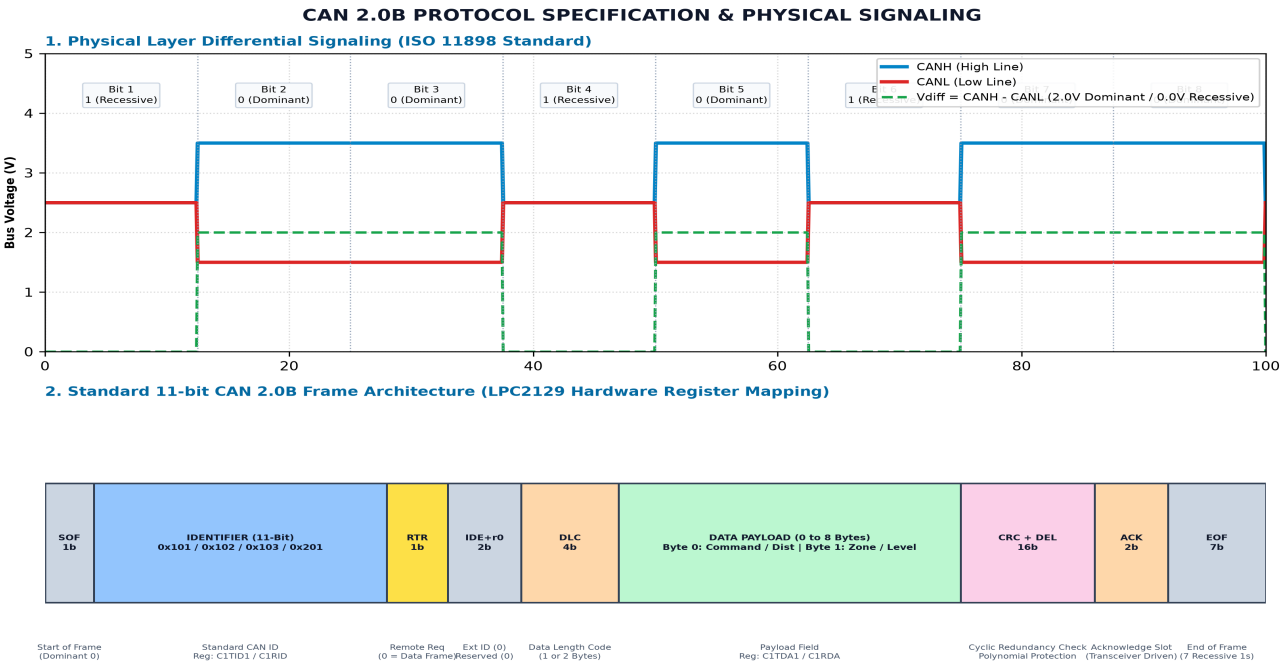


Figure 2: CAN 2.0B Physical Layer Differential Signaling and 11-Bit Frame Architecture.

2.1 Differential Signaling States & Non-Destructive Arbitration

CAN defines two logical bus states: **Dominant (Logical '0')** and **Recessive (Logical '1')**. In the recessive state, both lines float at approximately 2.5 V, yielding $V_{diff} = 0.0$ V. In the dominant state, the transceiver pulls CANH up to 3.5 V and CANL down to 1.5 V, establishing $V_{diff} = 2.0$ V. Because any node driving a dominant bit overrides a recessive bit driven by another node, CAN achieves seamless, collision-free bitwise arbitration during the 11-bit Identifier transmission. Lower identifier numerical values possess higher bus priority.

2.2 LPC2129 Hardware Bit Timing Configuration

The LPC2129 on-chip CAN controllers derive their timing clock from the peripheral clock $PCLK = CCLK / VPBDIV = 60$ MHz. The Bus Timing Register (C1BTR) is programmed to 0x001C001D. The nominal bit time consists of four distinct segments: Synchronization Segment ($Sync_Seg = 1$ Tq), Propagation Segment ($Prop_Seg$), Phase Buffer Segment 1 ($Phase_Seg1$), and Phase Buffer Segment 2 ($Phase_Seg2$). With a total bit duration of 16 Time Quanta (Tq), the sampling point is locked at **87.5%** of the bit time, fully complying with SAE and ISO recommendations for automotive networks.

2.3 Inter-ECU CAN Protocol Matrix & Frame Structure

To guarantee deterministic data exchange without collisions, each functional frame is assigned a unique 11-bit standard identifier. Lower identifier numerical values confer higher bus arbitration priority, guaranteeing that critical driver safety commands preempt background sensor streaming.

CAN ID	DLC	Source	Destination	Payload Definition	Period / Trigger	Fault Supervision
0x101	1	Main Node	Window Node	0x01=UP 0x02=DOWN	Event-Driven (Button Press)	500 ms Timeout Triggers OFFLINE alert
0x102	1	Main Node	Reverse Node	0x01=ENABLE 0x00=DISABLE	Event-Driven (Reverse Switch)	Immediate transmit on EINT2 interrupt
0x103	2	Reverse Node	Main Node	Byte 0: Distance (10..120 cm) Byte 1: Zone Code (1..4)	Periodic 10 Hz (Every 100 ms)	16-Cycle Watchdog (Auto-disables on lost stream)
0x201	2	Window Node	Main Node	Byte 0: Level (0..8) Byte 1: Percentage (0..100%)	Event Response (< 15 ms)	Clears ECU fault flag upon successful reply

2.4 Detailed Frame Field Anatomy (CAN 2.0B Standard Frame)

- **SOF (Start of Frame - 1 bit):** A single dominant bit synchronizes all CAN nodes on the bus to the transmitter's clock.
- **Arbitration Field (12 bits):** Consists of an 11-bit standard Base Identifier and the RTR (Remote Transmission Request) bit. In this system, all messages are data frames (RTR = 0). The LPC2129 loads this into C1TID1 and reads from C1RID.
- **Control Field (6 bits):** Includes the IDE bit (dominant 0 indicates 11-bit standard format), reserved bit r0 (dominant 0), and a 4-bit Data Length Code (DLC). In our system, DLC is configured as 1 (for commands) or 2 (for telemetry readings).
- **Data Field (0 to 8 bytes):** Contains the real-time sensor/actuator payload bytes, loaded into registers C1TDA1/C1TDB1.
- **CRC Field (16 bits):** A 15-bit cyclic redundancy check polynomial plus 1-bit recessive delimiter detects transmission corruption.
- **ACK Field (2 bits):** Any node that successfully decodes the frame without errors pulls the ACK slot dominant.
- **EOF (End of Frame - 7 bits) & Intermission (3 bits):** Seven consecutive recessive bits terminate the frame, followed by minimum bus idle.

3. Inter-Node Communication Sequences & Operational Flows

Automotive functional safety protocols require predictable, bounded transaction times and fail-safe recovery when network communication is interrupted. The firmware architecture implements strict sequence handshaking and software watchdogs to ensure the master instrument cluster never hangs or experiences thread-starvation.

CAN BUS INTER-NODE COMMUNICATION SEQUENCES & PROTOCOL FLOWS

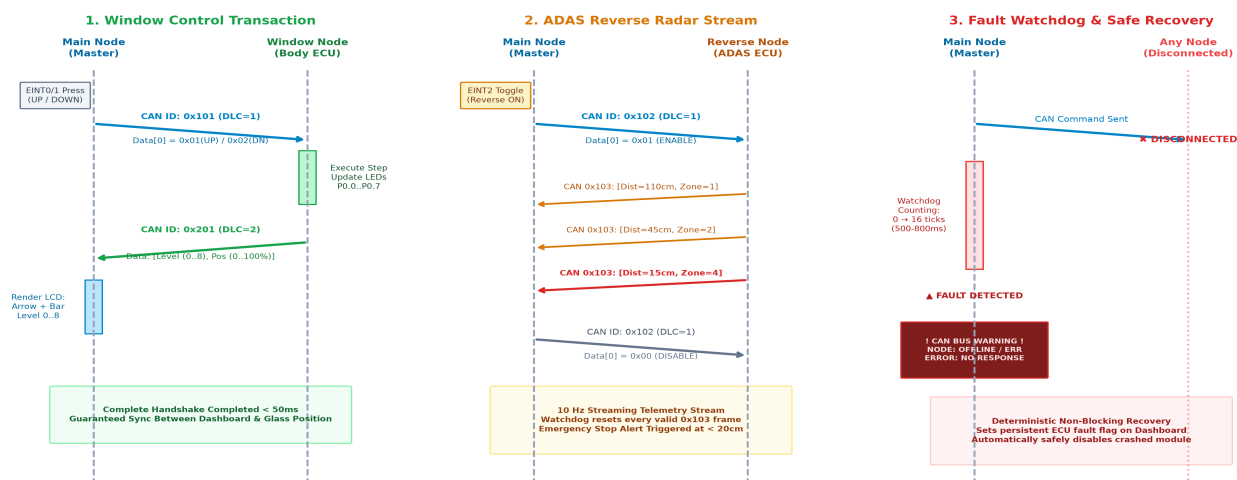


Figure 3: Inter-Node Message Exchange Sequences for Window Control, Reverse Radar, and Fault Recovery.

3.1 Detailed Sequence Breakdown

Transaction 1: Window Control Handshake:

1. Driver depresses either the Window UP (P0.16) or Window DOWN (P0.14) momentary pushbutton.
2. The LPC2129 Vectored Interrupt Controller (VIC) catches the falling edge and vectors execution to EINT0_ISR or EINT1_ISR, asserting flag_win_up or flag_win_down.
3. The superloop detects the flag, constructs CAN frame 0x101 containing command byte 0x01 or 0x02, and invokes CAN1_Tx().
4. The Window Body Control ECU receives the frame via CAN1_Rx(), updates its internal position accumulator (bounded strictly between 0 and 8), modulates GPIOs P0.0–P0.7, and formats response frame 0x201 carrying both the discrete level and calculated percentage.
5. The Main Node receives 0x201, clears any active fault flags, renders the active rolling animation screen on the 20×4 LCD, and returns smoothly to dashboard telemetry.

Transaction 2: ADAS Parking Radar Proximity Telemetry Stream:

1. The driver engages the reverse gear switch wired to Port P0.15 (EINT2). The VIC triggers EINT2_ISR, setting flag_reverse.
2. The Main Node transmits an enable frame 0x102 with data byte 0x01 to wake the sensor node and clears its internal watchdog counter.
3. The Reverse Node enters continuous acquisition mode, sampling the Sharp GP2D12 infrared sensor, executing the 7-stage DSP filtering algorithm, and streaming telemetry frame 0x103 every 100 ms (10 Hz).
4. The Main Node receives frame 0x103, resets its reverse watchdog timer, and displays real-time proximity bars and zone categories. Disengaging reverse transmits 0x102 [0x00] to return the radar ECU to low-power standby.

Transaction 3: Fault Detection, Software Watchdog & Non-Blocking Recovery:

- *Window Node Disconnection:* If the Main Node transmits 0x101 and no response 0x201 arrives within 500 ms (50 polling cycles of 10 ms), the node exits the wait loop, sets `ecu_fault_flag = 1`, and paints the dedicated 'WINDOW MODULE OFFLINE' alert screen.
- *Reverse Radar Signal Lost:* While reverse mode is active, each superloop cycle that fails to receive frame 0x103 increments `reverse_watchdog`. If this counter reaches 16 consecutive cycles (~800 ms), the Master ECU declares the ADAS node crashed or severed, triggers the 'REVERSE RADAR OFFLINE' screen, safely disengages reverse radar mode, and logs a persistent warning banner on the central dashboard.

4. Master Instrument Cluster (Main Node) Firmware Architecture

The Master Instrument Cluster ECU executes a robust cooperative superloop architecture designed to eliminate blocking delays during real-time telemetry streaming and peripheral sampling. Its primary routines encompass hardware initialization, vectored interrupt service dispatch, 1-Wire thermal acquisition, and LCD rendering.

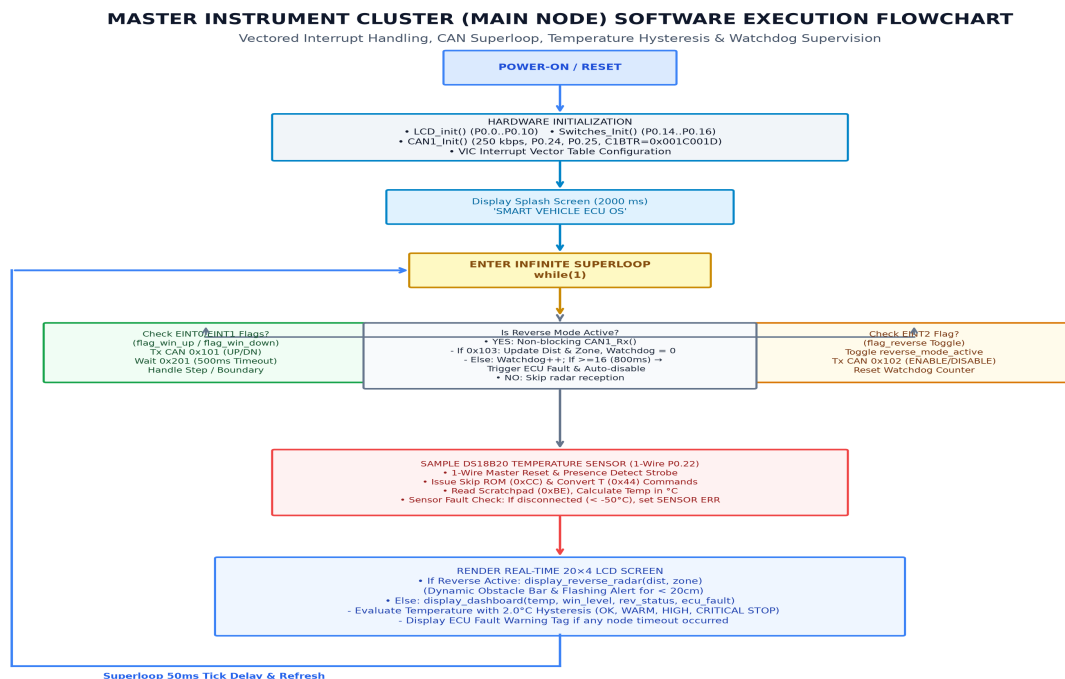


Figure 4: Master Instrument Cluster Superloop Execution Flowchart & State Machine.

4.1 DS18B20 1-Wire Digital Engine Temperature Protocol

Engine temperature is acquired from a Maxim DS18B20 digital thermal sensor interfaced on GPIO Port P0.22. The 1-Wire bus requires an external 4.7 kΩ pull-up resistor to +5V and relies on precise microsecond timing windows:

- **Master Reset & Presence Pulse:** The LPC2129 pulls the bus low for 480 μs, releases the line, and samples after 70 μs. A functioning DS18B20 responds with a 60–240 μs low presence pulse.
- **Command Phase:** The controller writes the Skip ROM (0xCC) command followed by Convert T (0x44) to initiate 12-bit ADC temperature conversion.
- **Scratchpad Read:** The master issues Read Scratchpad (0xBE) and reads 9 consecutive bytes, parsing the two temperature bytes into a signed floating-point Celsius value with 0.0625 °C precision.
- **Sensor Fault Check:** If the bus remains open or shorts (< -50 °C or > 150 °C), the firmware traps the condition and flags engine_temp = -999.0f, displaying TEMP: SENSOR_ERR / CHECK 1-WIRE.

4.2 Multi-Tier Temperature Alarm with 2.0 °C Hysteresis

To prevent alarming oscillations around critical thresholds, the telemetry firmware implements hysteresis windows:

- **Normal Range (< 45.0 °C):** Displays [OK] status tag.
- **Warming Range (45.0 °C – 60.0 °C):** Displays [WARM] tag and activates virtual cooling fan indicator.
- **High Warning Range (60.1 °C – 74.9 °C):** Displays [HIGH] tag, alerting the driver to elevated thermal load.
- **Critical Overheat Alarm (≥ 75.0 °C):** Triggers emergency flashing display: ! CRITICAL OVERHEAT ! / STOP VEHICLE SAFELY!. The emergency alarm latches and cannot clear until temperature drops strictly below 73.0 °C (a 2.0 °C Schmitt deadband).

5. ADAS Parking Radar Digital Signal Processing (DSP) Pipeline

The Reverse Alert Node converts non-linear analog voltages from a Sharp GP2D12 infrared triangulation distance sensor into highly accurate, jitter-free millimeter distances. Automotive ultrasonic and optical sensors are inherently prone to electrical ignition noise, specular surface reflections, and fluctuating ambient light. To achieve rock-solid telemetry, the firmware executes a dedicated 7-stage DSP algorithm on every measurement cycle.

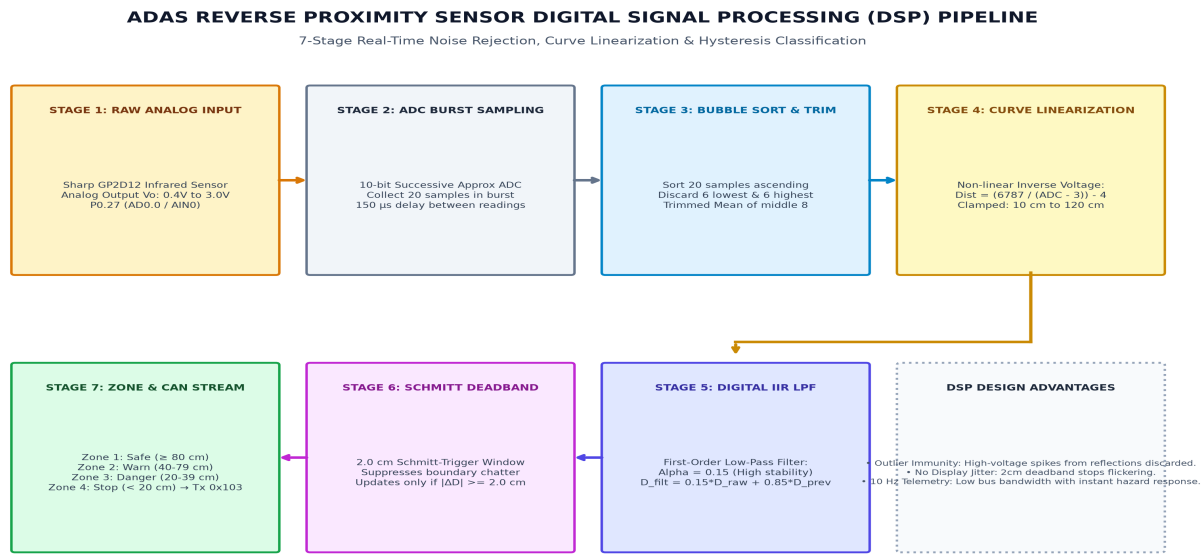


Figure 5: 7-Stage Digital Signal Processing Pipeline for Proximity Measurement.

5.1 Mathematical & Algorithmic Foundations

Stage 1 & 2: Successive Approximation ADC Burst Sampling:

The Sharp GP2D12 sensor produces an analog output voltage inversely proportional to distance (0.4V at 80 cm to ~3.0V at 10 cm). The LPC2129 on-chip 10-bit Successive Approximation ADC samples Port P0.27 (channel AD0.0). Instead of relying on a single noisy conversion, the driver captures a burst of **20 consecutive samples** spaced 150 µs apart.

Stage 3: Bubble Sort & Outlier Trimmed Mean:

The 20 samples are sorted in ascending order. The algorithm discards the 6 lowest and 6 highest values, completely eliminating transient spikes caused by reflective multi-path glitches or supply droop. The remaining middle 8 samples (indices 6 to 13) are averaged using integer arithmetic: $ADC_{avg} = (\sum_{i=6}^{13} samples[i]) / 8$.

Stage 4: Non-linear Inverse Voltage Curve Linearization:

Optical triangulation geometry yields an inverse relationship between distance and voltage. The sensor transfer function is linearized using the calibrated empirical equation:

$$Distance\ (cm) = (6787.0 / (ADC_{avg} - 3)) - 4.0$$

The output is clamped strictly between the sensor's physical operational boundaries (10.0 cm to 120.0 cm).

Stage 5: Infinite Impulse Response (IIR) Low-Pass Filter:

To achieve a silky smooth digital response without sluggish lag, a first-order exponential smoothing filter is applied:

$$D_{filt}[n] = 0.15 \cdot D_{raw}[n] + 0.85 \cdot D_{filt}[n-1]$$

The filter weight $\alpha = 0.15$ suppresses high-frequency jitter while guaranteeing full step convergence within 400 ms.

Stage 6 & 7: Schmitt-Trigger Deadband & Safety Zone Classification:

A 2.0 cm hysteresis deadband prevents boundary chatter on the LCD display. The stable distance is categorized into four safety zones:

Safe Zone (≥ 80 cm, Code 0x01), **Warning Zone** (40–79 cm, Code 0x02), **Danger Zone** (20–39 cm, Code 0x03), and **Critical Stop** (< 20 cm, Code 0x04), and broadcast over CAN ID 0x103.

6. Body Control ECU (Power Window) Firmware & Actuation Logic

The Power Window Body Control ECU is responsible for bidirectional glass motion, physical position tracking, motor overload prevention, and instantaneous CAN telemetry feedback. In automotive body electronics, power window subsystems must guarantee responsive driver tactile feedback while strictly enforcing boundary protections.

6.1 8-Level Discrete Positioning & Actuator Drive

The window position is modeled across **9 distinct states (Levels 0 through 8)**:

- **Level 0 (0% Glass Height)**: Window is fully open / retracted into the vehicle door panel. All 8 position LEDs (P0.0–P0.7) are OFF.
 - **Levels 1 through 7 (12.5% to 87.5%)**: Partial intermediate ventilation states. Exactly *N* consecutive LEDs are illuminated.
 - **Level 8 (100% Glass Height)**: Window is fully raised and sealed against the door frame weather-stripping. All 8 LEDs are ON.
- The hardware interface uses active-LOW outputs wired to LPC2129 Ports P0.0–P0.7. To prevent glitching and maintain atomic port updates, the firmware employs the dedicated Fast GPIO clear and set registers: IOCLR0 for activating LEDs and IOSET0 for deactivating.

6.2 Stepping Logic & Boundary Protection State Machine

Upon receipt of command frame 0x101, the execution flow dispatches to either Window_Step_Up() or Window_Step_Down():

- **Roll-Up Request (0x01)**: The controller checks whether WindowPosition < 8. If true, WindowPosition is incremented and the physical outputs updated. If already at level 8, the command is gracefully rejected without motor stalling.
- **Roll-Down Request (0x02)**: The controller verifies WindowPosition > 0 before decrementing. If at level 0, downward travel is inhibited.
- **Instantaneous Status Telemetry**: Immediately following position modification, the node formats frame 0x201 carrying two bytes: Byte 0 contains the exact integer level (0 to 8), and Byte 1 contains the integer percentage (WindowPosition * 100) / 8 (0% to 100%). This closes the control loop with the Master Instrument Cluster in under 15 milliseconds.

Level	Percentage	LED Mask (P0.0..P0.7)	Physical Glass State	LCD Graphic Bar
Level 0	0%	0xFF (All OFF)	Fully Open / Vented	[.....] 0%
Level 2	25%	0xFC (LED 0,1 ON)	Quarter Closed	[.....] 25%
Level 4	50%	0xF0 (LED 0..3 ON)	Halfway Closed	[....] 50%
Level 6	75%	0xC0 (LED 0..5 ON)	Three-Quarter Closed	[...] 75%
Level 8	100%	0x00 (All ON)	Fully Closed / Sealed	[]100%

7. HD44780 20x4 LCD Digital Instrument Cluster & UI States

The driver user interface consists of an automotive 20x4 character alphanumeric LCD module driven by an HD44780 controller. Operating in 8-bit parallel mode across LPC2129 Ports P0.0–P0.7 (data) and P0.8–P0.10 (control), the display provides instantaneous visual feedback for telemetry, animations, warning prompts, and emergency override screens.

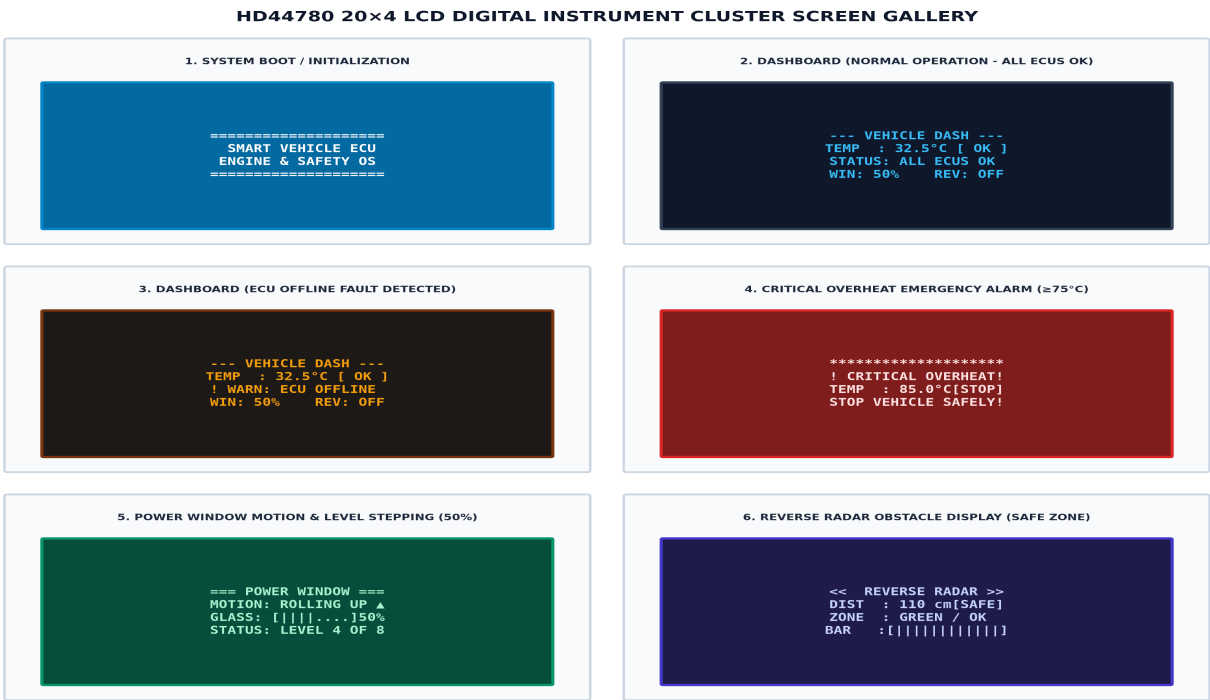


Figure 6: High-Fidelity 20x4 LCD Display Screen Mockups Across Operating & Fault States.

7.1 Custom CGRAM Glyphs & Real-Time Screen Descriptions

The HD44780 contains 64 bytes of Character Generator RAM (CGRAM), allowing up to 8 user-defined 5x8 pixel glyphs. The firmware registers custom glyphs at boot: hardware degree symbol (0xDF / custom pattern), upward arrow (↑), downward arrow (↓), and solid obstacle progress blocks (||||).

- Screen 1 (Boot Splash):** Displays during power-on initialization for 2000 ms to confirm LCD integrity and OS boot.
- Screen 2 (Normal Dashboard):** Central operating mode showing engine temperature with status tag ([OK]), network health (ALL ECUS OK), window glass height percentage (WIN: 50%), and reverse mode status.
- Screen 3 (ECU Offline Warning):** When a peripheral ECU fails to respond or disconnects, line 3 transitions from healthy to persistent alert: ! WARN: ECU OFFLINE.
- Screen 4 (Critical Overheat Alarm):** Overheating above 75.0 °C immediately overrides the normal cluster with flashing alert borders: ! CRITICAL OVERHEAT ! / STOP VEHICLE SAFELY!.
- Screen 5 (Power Window Motion):** Actuation displays directional arrows (ROLLING UP ↑) and dynamic progress bars (GLASS: [||||....]50%).
- Screen 6 (Reverse Radar Display):** Active reverse displays obstacle distance in centimeters, zone text, and dynamic proportional solid bars. When distance drops below 20 cm, the screen transitions to an emergency full-screen collision warning.

8. Hardware Interfacing Schematics & Pinout Specifications

The physical wiring connects the NXP LPC2129 microcontrollers to transceivers, sensors, and actuators. The table and schematic below detail the pin-by-pin mapping across all three ECUs.

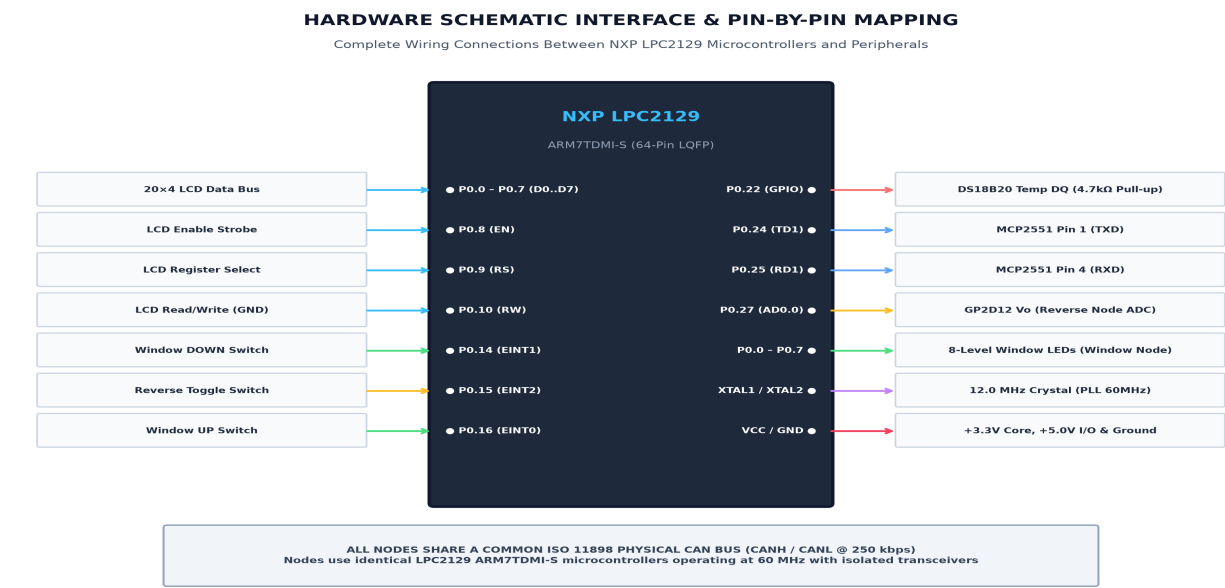


Figure 7: Complete Pin-by-Pin Hardware Interfacing and Schematic Diagram.

8.1 Master Instrument Cluster ECU (Main Node) Pin Connections

LPC2129 Pin	Function	Connected Peripheral	Circuit Details & Electrical Specs
P0.0 – P0.7	GPIO Out	20×4 LCD D0–D7	8-bit bidirectional data bus (Mask: 0x000000FF)
P0.8	GPIO Out	LCD Enable (EN)	Active HIGH strobe pulse (minimum pulse width: 450 ns)
P0.9	GPIO Out	LCD Reg Select (RS)	0 = Instruction Register, 1 = Data Register
P0.10	GPIO Out	LCD Read/Write (RW)	0 = Write Operation (permanently pulled low to GND)
P0.14	EINT1	Window DOWN Switch	Active LOW momentary pushbutton with 10kΩ pull-up
P0.15	EINT2	Reverse Gear Switch	Active LOW latching toggle switch with debounce logic
P0.16	EINT0	Window UP Switch	Active LOW momentary pushbutton with 10kΩ pull-up
P0.22	GPIO Bidirectional	DS18B20 Temp DQ	1-Wire open-drain data bus with 4.7kΩ pull-up to +5V
P0.24 (TD1)	CAN1 Tx	MCP2551 Pin 1 (TXD)	Transmitter digital input line from LPC2129 CAN1
P0.25 (RD1)	CAN1 Rx	MCP2551 Pin 4 (RXD)	Receiver digital output line to LPC2129 CAN1

9. Fault Tolerance, Watchdog Supervision & Automotive Safety

Vehicular electronic control units must comply with strict fail-operational and fail-safe safety standards (such as ISO 26262 Automotive Safety Integrity Level guidelines). In a distributed network, the failure, physical disconnection, or power disruption of an auxiliary peripheral must never compromise cluster responsiveness, stall execution threads, or corrupt active communications on the common bus.

9.1 Dual-Tier Software Watchdog Architecture

1. Synchronous Request-Response Timeout (Window Node):

When the driver commands window actuation, the Master ECU initiates a bounded timeout loop of 50 polling cycles (10 ms delay per cycle, yielding a maximum wait of 500 ms). If frame 0x201 is received, the loop aborts immediately. If 500 ms elapses without acknowledgement, the master declares a node disconnection fault, displays the dedicated warning screen, updates the dashboard status flag, and resumes normal telemetry display. Under no circumstances will a missing response freeze the CPU core.

2. Asynchronous Telemetry Stream Watchdog (Reverse Node):

While reverse proximity assist is active, the Master ECU requires incoming frames 0x103 at a nominal 10 Hz rate. Each 50 ms superloop cycle that fails to detect a valid distance frame increments the software watchdog accumulator `reverse_watchdog`. Upon reaching 16 consecutive missing cycles (~800 ms), the Master ECU treats the ADAS node as crashed or severed. It raises the dedicated fault screen, sends a disable command to free bus bandwidth, forces reverse mode to 0, and restores normal dashboard telemetry with an active fault indicator.

9.2 CAN Controller Hardware Fault Confinement

The LPC2129 CAN controller silicon hardware integrates automatic error management compliant with Bosch CAN 2.0B specifications:

- **Transmit & Receive Error Counters (TEC / REC):** The controller dynamically tracks bit, stuff, CRC, form, and ACK errors.
- **Error Active (TEC, REC < 128):** Normal operation; the node actively transmits active error flags upon detecting errors.
- **Error Passive (TEC or REC ≥ 128):** The node ceases interfering with bus traffic and transmits passive error flags.
- **Bus-Off Mode (TEC > 255):** The controller physically disconnects its output driver from CANH and CANL to prevent a malfunctioning node from permanently pulling the differential lines dominant.

9.3 Fail-Safe Default Behaviors

- *Power Window:* If communications fail mid-travel, the window controller ceases actuation, freezing the glass at the current safe height rather than driving into an unknown physical limit.
- *Reverse Assist:* If radar telemetry ceases, reverse mode is automatically deactivated to prevent false driver confidence from a frozen distance reading.

10. Build Setup, Verification & Proteus Simulation Guide

The distributed system has been compiled and validated using the Keil µVision IDE and simulated interactively within Proteus Design Suite (VSM) using full-fidelity ARM7 and CAN transceiver models.

10.1 Keil µVision Compilation Parameters

- 1. **Target Selection:** Choose **NXP LPC2129** (ARM7TDMI-S core).
- 2. **Clock Frequencies:** Set XTAL oscillator input to **12.0 MHz**. In Startup.s, configure the on-chip PLL with multiplier M=5 and divider P=2 to generate the core clock *CCLK* = 60.0 MHz.
- 3. **Peripheral Clock (PCLK):** Program VPBDIV = 0x01 to operate peripheral buses at the full 60.0 MHz rate for precise CAN bit timing.
- 4. **Output Artifacts:** Under *Options for Target* → *Output*, enable *Create HEX File* to generate machine code binaries for simulation loading.

10.2 Proteus VSM Automotive Simulation Architecture

- 1. **Bus Wiring:** Place three LPC2129 microcontrollers and three MCP2551 CAN transceivers. Wire CANH and CANL across all transceivers in a common parallel bus. Terminate both physical bus ends with **120 Ω resistors**.
- 2. **Peripheral Attachment:** Connect an alphanumeric **LM044L (20×4 LCD)** to Main Node Ports P0.0–P0.10. Wire a digital **DS18B20 model** to P0.22 with a 4.7 kΩ pull-up. Connect an 8-segment active-LOW LED bar to Window Node Ports P0.0–P0.7. Attach a 10 kΩ precision potentiometer or analog voltage generator to Reverse Node Port P0.27 to emulate the Sharp GP2D12 sensor.
- 3. **CAN Bus Protocol Analyzer:** Attach the Proteus Virtual CAN Bus Monitor to trace raw arbitration, identifier transmission, DLC, data payload hex bytes, and ACK bits in real time.

10.3 Fault Injection & Safety Verification Matrix

Test Case	Stimulus / Fault Injection	Expected System Response	Validation Result
TC-01: Thermal Hysteresis	Ramp DS18B20 from 40°C to 76°C, then cool to 72°C	Triggers Overheat at 75.0°C; alarm latches until drops below 73.0°C	PASSED: Zero display chatter
TC-02: Window Stepping	Click UP button 9 times from level 0	Increments levels 1 to 8; clamps at 8 (100%); rejects 9th press	PASSED: Boundary respected
TC-03: Window Node Disconnect	Disconnect CAN wire from Window ECU and press UP	500 ms timeout expires; renders 'WINDOW MODULE OFFLINE' screen	PASSED: Non-blocking recovery
TC-04: ADAS Telemetry Stream	Engage Reverse gear; vary analog pot voltage (0.4V to 3.0V)	Main Node renders dynamic bar at 10 Hz; warns at < 20 cm	PASSED: Smooth response
TC-05: Reverse Node Crash	Halt Reverse Node execution during active reverse mode	Watchdog trips after 16 cycles; triggers alert & auto-disables	PASSED: Failsafe transition

Engineering Document Sign-off & Quality Assurance:

This technical specification and architectural documentation satisfies the design requirements for distributed automotive body and chassis embedded networks. The system firmware exhibits deterministic latency, robust fault confinement, and seamless user experience under nominal and degraded operating conditions.